GE Healthcare

Technical Publication

Direction 5174051-100
Revision 2

2 Phase UPS Installation

For use with
BrightSpeed,
LightSpeed (1.x- 5.x) &
LightSpeed Pro 16 Systems

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IMPORTANT PRECAUTIONS

Language

WARNING	▪ This Service Manual is available in English only. ▪ If a customer's service provider requires a language other than English, it is the customer's responsibility to provide translation services. ▪ Do not attempt to service the equipment unless this service manual has been consulted and is understood. ▪ Failure to heed this Warning may result in injury to the service provider, operator or patient from electric shock, mechanical or other hazards.
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All packages should be closely examined at time of delivery. If damage is apparent, have notation "damage in shipment" written on all copies of the freight or express bill before delivery is accepted or "signed for" by a General Electric representative or a hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

To file a report:

Call 1-800-548-3366 and use option 8.

Fill out a report on http://us44hdd21/sctq/InstallFulfill/InstalFulfillment.htm Contact your local service coordinator for more information on this process.

Rev. Jan. 5, 2005

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

OMISSIONS & ERRORS

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GE personnel, please use the GE Healthcare PQR Process to report all omissions, errors, and defects in this publication.

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REVISON HISTORY

Revision	Date	Reason for change
Rev A	04/11/2006	Initial Draft
Rev 1	17 May 2006	Release (Including validation feedback)
Rev 2	21 Nov 2006	PQR 13086320: removed Chapter 6 schematic.

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Chapter 1
Common Instructions and Requirements

Section 1 - Scope

This installation specification is a control document for the installation of an Uninterruptible Power System (UPS) unit (B7999ZA) to be used with General Electric Healthcare (GEHC) Computerized Tomography diagnostic imaging systems. This document supports CT Systems, which use the following Power Distribution Units.

NGPDU models:

- 2326492
- 2326492-2
- 2326492-3

CT Compact PDU Models:

- 2133533
- 2269902

It also includes the instructions for interconnection with the system A1 Main Disconnect & UPS Control Panel.

Section 2 - Overview

This document was designed to guide installation of a 10kVA UPS into BrightSpeed, LightSpeed, or LightSpeed Pro [16] systems using an NGPDU. However, backward compatibility with older systems utilizing CT Compact PDUs may be achieved by following the details shown in the attached schematics.

The UPS powers critical system computer and control functions. It is intended to bridge short power outages and provide time for crossover from normal mains power to emergency power. It is sized to provide at least 10 minutes backup with fully charged batteries.

It is recommended that if a power outage exceeds 5 minutes with no mains or emergency power, the patient should be removed from the table and an orderly shutdown of the system should be initiated.

For safe and orderly system start-up & shutdown this UPS kit must be installed with a properly configured A1 Main Disconnect Panel. Use of one of the following GE panels, or equal, specifically designed for use with this kit, is recommended:

- E4502AB / P5051RJ (90A)
- E4502AC / P5051RK (110A)
- E4502AE (125A)
- E4502AF (150A)

Refer to your system Pre-Installation manual for proper selection.

2.1 Tools Required

- FE Tool Kit
- LOTO
- Fluke 87 Multi-meter or equal
- Wire Stripper
- Labels

2.2 Estimated Manpower & Time Requirements

- 1 FE & 1 Electrician
- 3 hours

Section 3 - Safety Requirements

General Warning Considerations

The only user operations permitted with the UPS are:

- Starting up and shutting down of the UPS.
- Operating the user interface in the UPS.
- Connecting data interface cables.
- Monitoring the UPS.

Note: It is strongly recommended to have the User's Guide for the UPS available. These operations must be performed according to the instructions in this document and the User's Guide for the UPS being installed.

This document covers installation of the POWERWARE **9155-10GE** UPS only.

DANGER

THE UPS CONTAINS LETHAL VOLTAGES. SERVICE AND REPAIRS SHOULD BE PERFORMED BY AUTHORIZED PERSONNEL ONLY. THERE ARE NO USER SERVICEABLE PARTS INSIDE THE UPS.

Section 4 - Preliminary Instructions

4.1 Electrician

An electrician will be required to install a 3/4" conduit between the A1 Disconnect and the computer floor or to the general area of the UPS. The conduit should be terminated in a small wall box. This will be used for the UPS Control cable connecting the UPS and A1 Main Disconnect Panel.

4.2 Electrical Requirements

The UPS receives power from the PDU and returns power to the PDU. No other power connections are required. However, low voltage and 120Vac SEO Control and interlock signals are passed between the UPS and A1 Panel.

Chapter 2
Powerware 9155-10GE UPS Installation

Perform the following installations and connections at the UPS.

Section 1 - UPS Mechanical Preparation

1.1 Site Preparation Verification

Verify the site has been prepared for the UPS installation. Locate the Powerware User's Guide supplied with the UPS kit and refer to Chapter 3 as necessary for additional information.

1.2 Inspection

The UPS is shipped bolted to a wooden pallet as shown below and protected with an outer protective wrap. Visually verify there is no evidence of shipping damage to the packaging or UPS. Remove the outer protective wrap and perform the steps in the following section to unload the UPS from the pallet.

Figure 1 - UPS on Shipping Pallet

1.3 Unloading the UPS from the Pallet

Note: The following information has been extracted from the Powerware User's Guide provided with the UPS. Refer to that document for additional information if necessary.

CAUTION

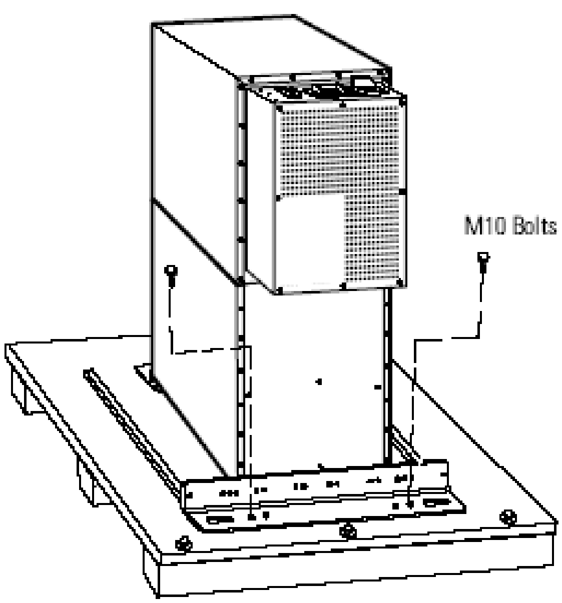
- The UPS cabinet is heavy. Shipping weight is approximately 170kg (375lb). If unpacking instructions are not closely followed, the cabinet may tip and cause serious injury.
- Do not install a damaged cabinet. Report any damage to the carrier and contact your Eaton service representative immediately.
- Do not tilt the UPS cabinet more than 10° from vertical or the cabinet may tip over.
- Do NOT use the maintenance bypass switch housing as a handle while removing the cabinet from the pallet.

To remove the pallet:

1. Remove the two M10 bolts securing the stabilizing bracket to the pallet
2. Remove the four M4 screws securing the stabilizing bracket to the cabinet rear panel and remove the bracket. Retain the hardware for later use.

NOTE Be sure to retain the stabilizing bracket and hardware for later reassembly onto the cabinet.

3. Remove the front cover from the bottom cabinet to access the front shipping bracket.. Press and release the handle latch at the bottom of the cover and then lift the cover up and off the cabinet.
4. Remove the three M10 bolts securing the rear shipping pad to the pallet and remove the shipping pad.
5. Remove the two M10 bolts securing the front shipping bracket and remove the bracket. If needed, adjust the leveling feet to release the bracket.



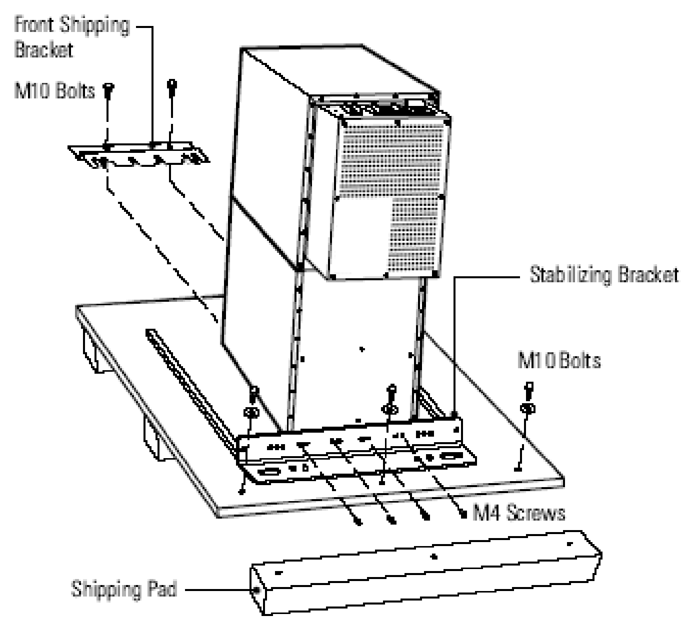


Figure 2 - Shipping Bracket Removal

6. Reinstall the front cover removed in Step 3. Hang the top edge of the cover on the cabinet first, then lower the bottom edge and snap into place.

NOTE *In the following steps the pallet tilts and acts as a ramp once the UPS is rolled beyond the center of the pallet. Be sure to restrain the UPS as it continues to roll down the pallet / ramp. The front shipping bracket acts as a brake to assist restraining the UPS.*

WARNING

Do not stand directly behind the pallet while unloading the cabinet. If unloading instructions are not closely followed, the UPS may roll unexpectedly and cause serious injury.

7. Slowly roll the cabinet toward the rear of the pallet. Once the pallet tilts, continue rolling the cabinet down the pallet until the cabinet touches the floor. If needed, adjust the leveling feet so that the cabinet will roll.
8. With the cabinet supported, slowly pull the pallet away from the cabinet.

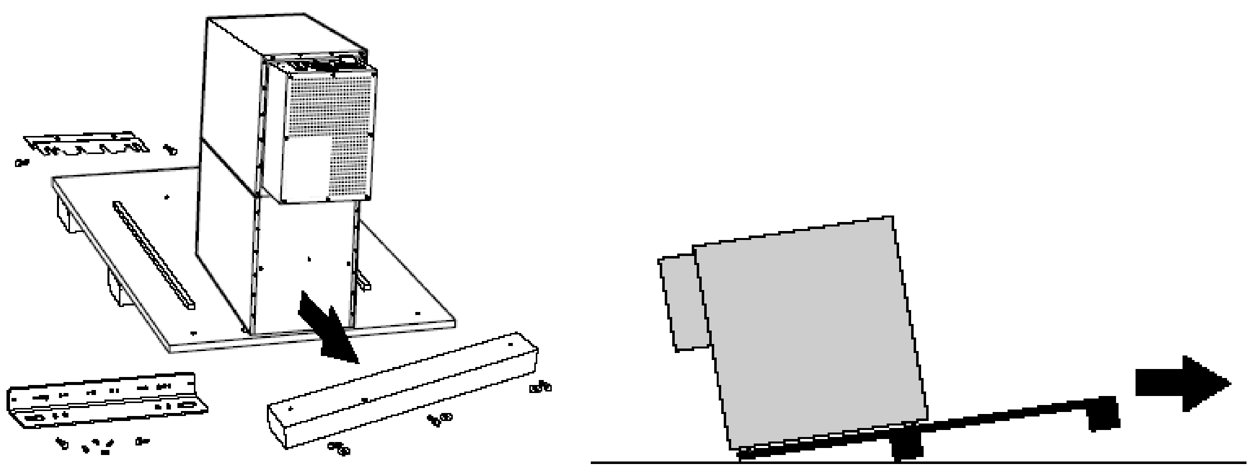


Figure 3 - Removing the Pallet

9. If installing the UPS permanently, retain the hardware for later use; otherwise, discard or recycle the pallet and hardware in a responsible manner.
10. Roll the UPS to its final installation location.
11. Secure it in position by lowering the leveling feet until the UPS is not resting on the casters and it is level.
12. If a more secure installation is desired, the retained hardware may be used to secure the UPS cabinet to the floor.
13. Reinstall the front and rear shipping brackets to the UPS with the angles facing outward.
14. Attach the angle brackets to the floor with contractor-supplied hardware.

1.4 Mobile Applications kit (B7999ZM) - Vibration Mount Assembly

When used in mobile applications the UPS is mounted to a vibration isolation system. The components used are supplied with the Mobile UPS kit, catalog # B7999ZM.

Perform the follow steps:

1. Raise the UPS to access the bottom of the unit.
2. Remove the four leveling feet from the bottom of the UPS.
3. Mount two isolator bars (5169868) to the base of the unit as shown below.

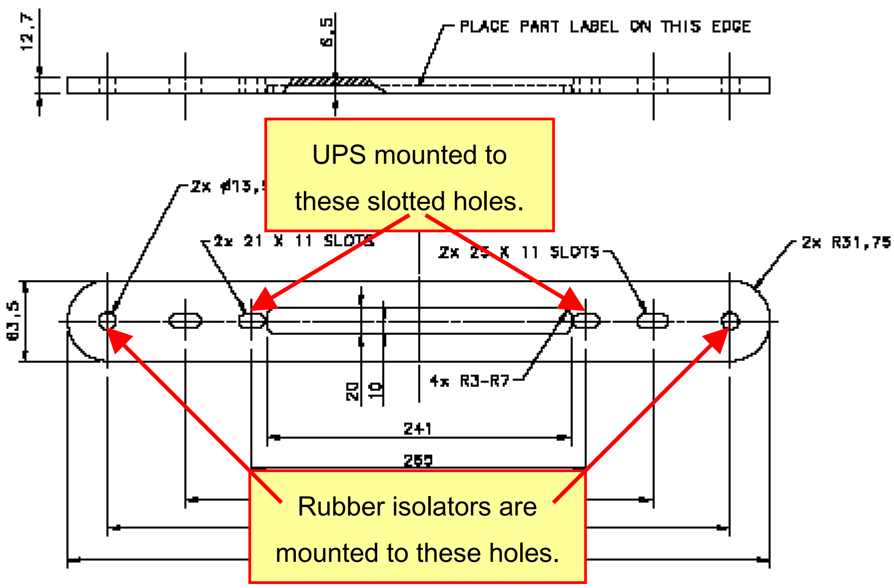


Figure 4 - Isolator Bars (5169868)

4. Locate the two inner slotted holes on the Isolator Bar.
5. Align the two slotted holes of the Isolator Bar to the existing holes on the bottom of the UPS where the leveling feet were removed.
6. Attach the Isolator Bar to the UPS using the hardware shown below & supplied with the Mobile UPS kit.

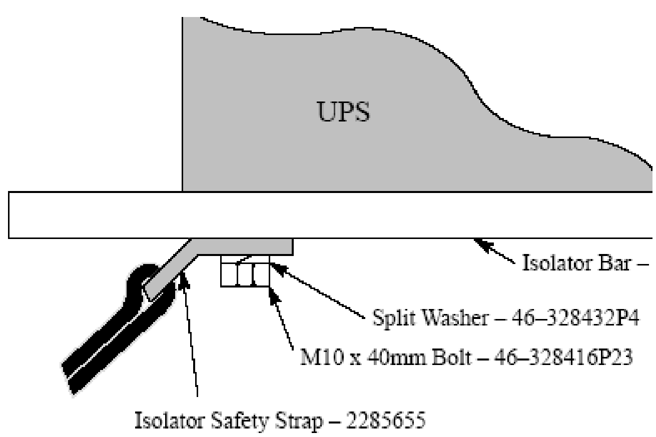


Figure 5 - Isolator Bar Mounting

7. Secure the Safety Strap and isolator bar to the bottom of the UPS using an M10 x 40 mm long Bolt – Part No. 46–328416P23 and a Split Washer – Part No. 46–328432P4.

NOTE: The Safety Strap is positioned to the side of the Isolator Bar as shown below.

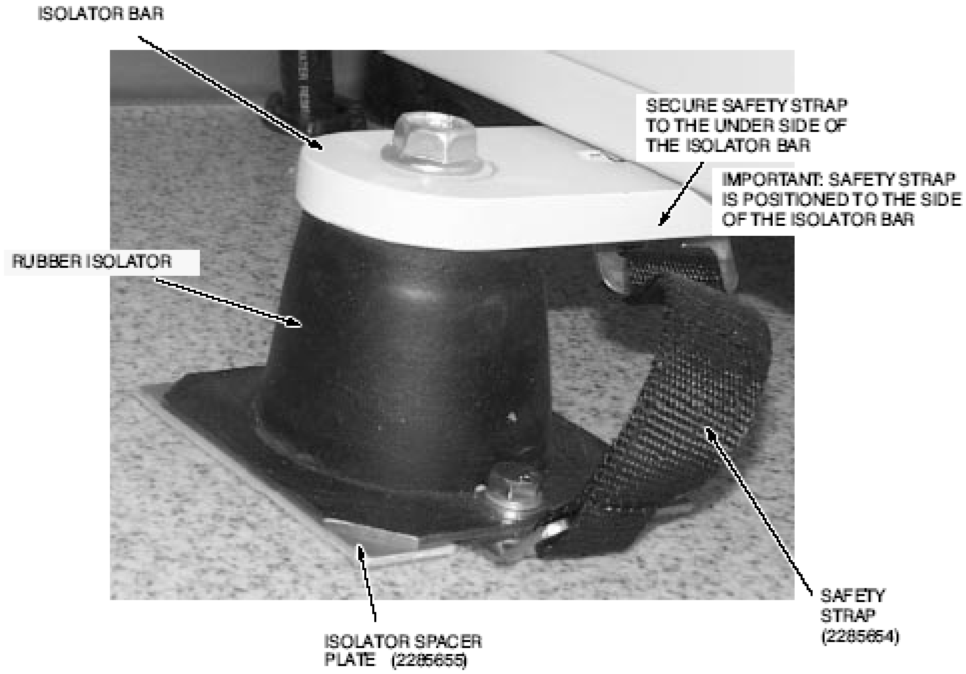


Figure 6 - Safety strap Installation

8. Rubber Isolators – Part No. 2282295 are installed at each end of the Isolator Bar.
9. Align the holes on the top of the Rubber Isolator to the outer holes on the ends of the Isolator Bars.
10. Position the Rubber Isolator mounting hole parallel to the UPS as shown above.
11. Secure using a 1/2–13 x 1–1/4" long Mounting Bolt, Part No. 46–221850P13.
12. Place Isolator Spacer Plate – Part No. 2285655 under each Rubber Isolator. Ensure that the cut out of each Isolator Spacer Plate is positioned towards the unsecured end of the Safety Strap and the Rubber Isolator mounting hole.

Figure 7 - Safety strap Alignment

13. Place unsecured end of Safety Strap into the cut out of the Isolator Spacer Plate.

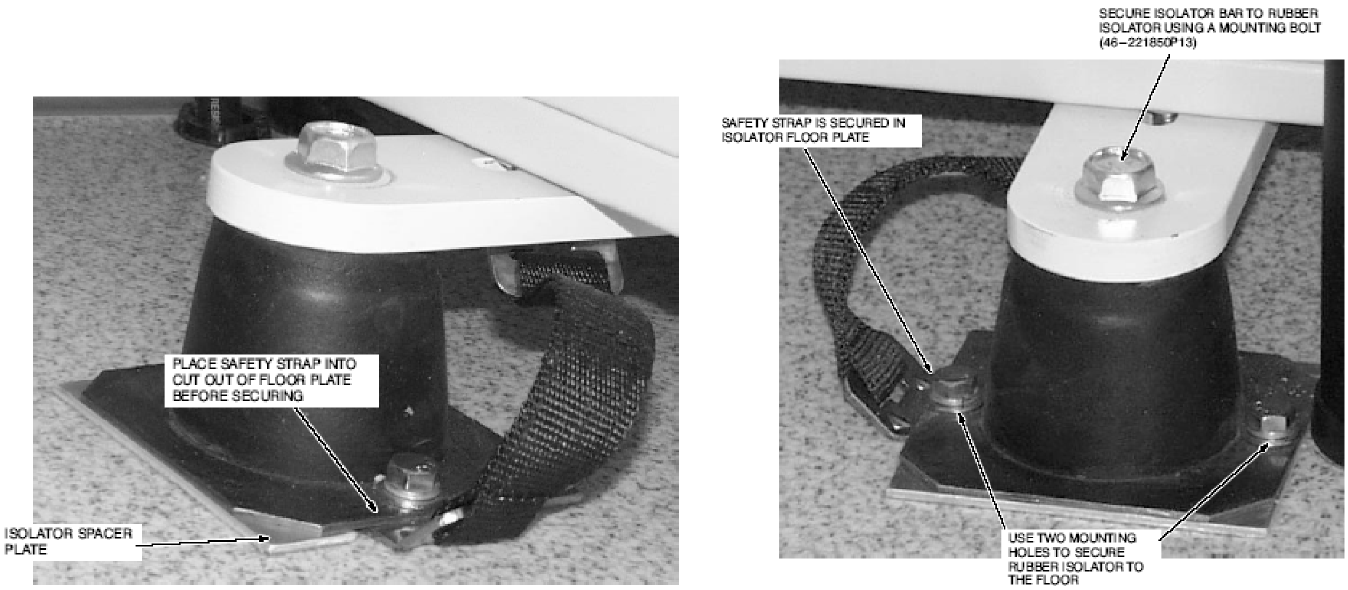


Figure 8 - Safety strap Alignment Details

14. Install each Rubber Isolator to the desired trailer floor location using the proper mounting hardware provided by the trailer vendor.
15. Insert the Safety Strap end under the Rubber Isolator and between the floor and the Isolator Spacer Plate cut out.

Section 2 - Powerware 9155-10GE UPS Connections

2.1 UPS I/O Terminal Compartment

All power and control connections are made in the I/O terminal compartment located on the rear of the UPS. To access the compartment, remove the eight machine screws holding the cover in place and set aside for later use.

1) Verify both the UPS battery and output breakers on the top rear panel are in the “OFF” position.
2) Set the manual bypass switch to the “SERVICE” position.
3) Remove the rear terminal box cover from the UPS.

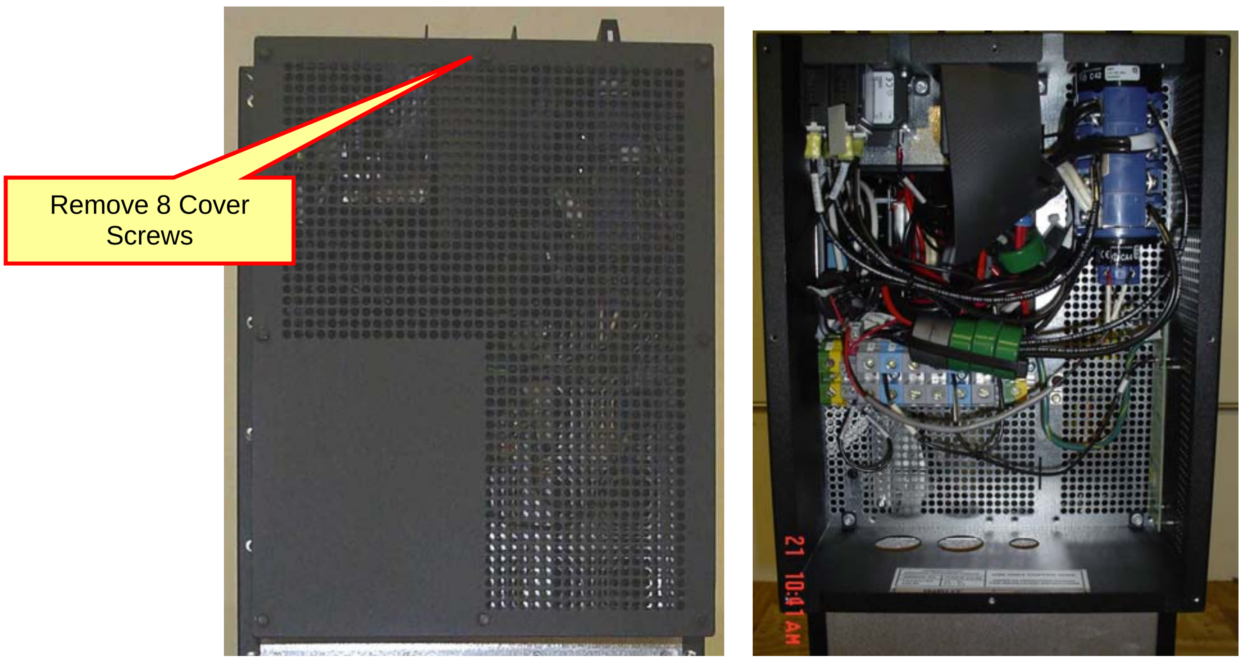


Figure 9 - UPS I/O Terminal Compartment

4) Confirm by measurement that all terminals on the terminal strip are at 0V with respect to chassis.
5) Install two 3/4” & one 1/2” cord grip fittings in rear of UPS as shown in Figure 9. Use reducer plates and hardware supplied in the kit as required. (Note: Enlarge small if required for cord grip installation.)

2.2 UPS Input Power Cable from PDU

Connect UPS Input Power Cable (P/N 5169294) as follows:

1) Pull cable end labeled “To UPS TB1” through the left most 3/4” cord grip in the back of the UPS, to TB1.
2) Connect the GREEN/YELLOW wire to “GND”.
3) Connect the BLACK wire to “L1”. Cable not marked as L1 it is marked as TBx
4) Connect the BLUE wire to “NEUT”. Cable not marked as L1 it is marked as TBx
5) Connect the ORANGE wire to “L2”. Cable not marked as L1 it is marked as TBx
6) If present, cut off RED wire at cable jacket.
7) Securely tighten the cord grip onto the input cable.

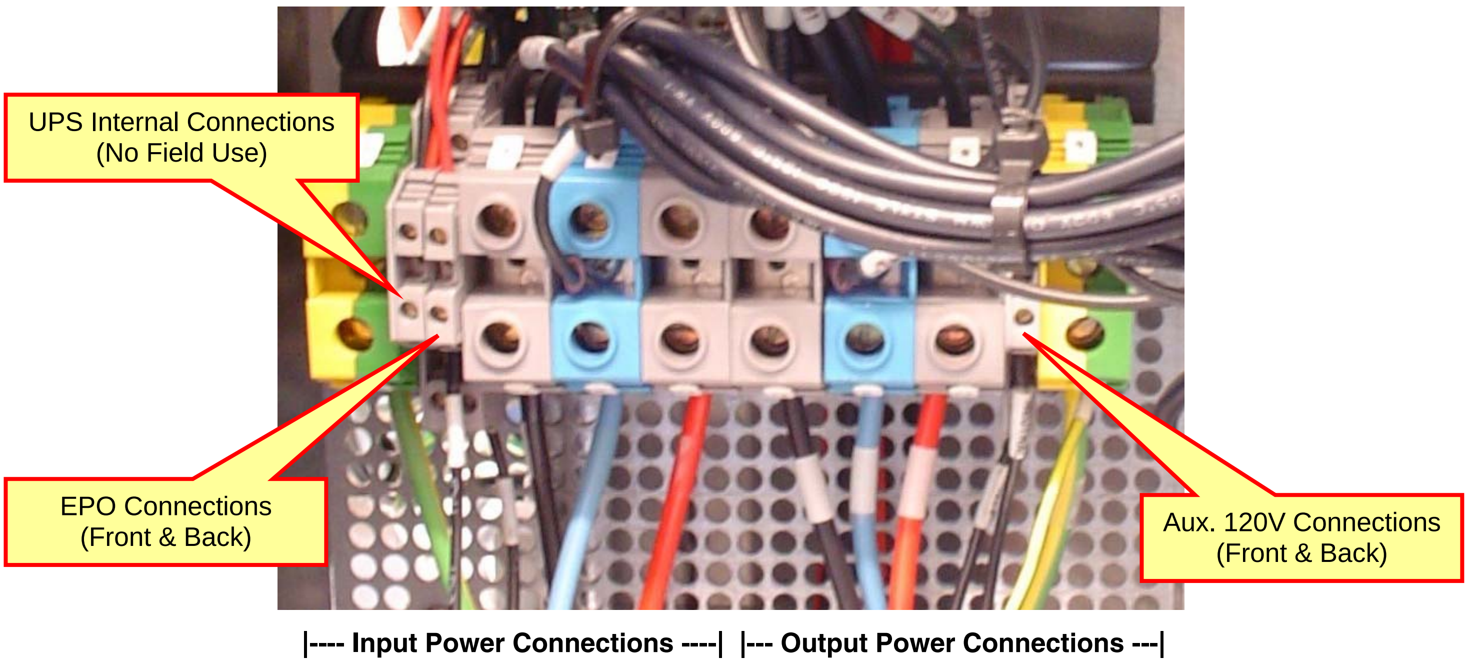


Figure 10 - UPS TB1 Connections

2.3 UPS Output Power Cable to PDU

Connect UPS Output Power Cable (P/N 5169294-2) as follows:

1) Pull cable end labeled “To UPS TB1” through the right 3/4” cord grip in the back of the UPS, to TB1.
2) Connect the BLACK wire to “L1”.
3) Connect the BLUE wire to “NEUT”.
4) Connect the ORANGE wire to “L2”.
5) Connect the GREEN/YELLOW wire to “GND”.
6) If present, cut off RED wire at cable jacket.
7) Securely tighten the cord grip onto the output cable.

2.4 UPS Control Cable to A1 Main Disconnect / UPS Control Panel

Referring to Figure 10 above, connect the UPS Control cable (P/N 5169224) as follows:

1) Route the cable end labeled “To UPS” through the 1/2” cord grip in the bottom of the UPS Terminal Box.
2) Connect the GREEN/YELLOW (GND) and SHIELD wires to the grounding terminal “GND”.
3) Remove the existing jumper connected between TB1-2A1 & 2A2, if present.
4) Connect the BLK 1 (EPO_1) wire to TB1-2A1.
5) Connect the BLK 2 (EPO_2) wire to TB1-2A2.
6) Connect the BRN (AUX 120) wire to TB1-8AL.
7) Connect the BLU (NEUT) wire to TB1-8AN.
8) Securely tighten the cord grip onto the control cable.

When all connections are secure, re-install the terminal box cover at the rear of the UPS.

Chapter 3
Power Distribution Unit

Section 1 - NGPDU Connections

This UPS kit has been specifically designed for compatibility with BrightSpeed, LightSpeed 5.x, or LightSpeed Pro [16] CT systems. These systems utilize an NGPDU. Model numbers currently in use are:

- 2326492
- 2326492-2
- 2326492-3

The same interconnection information applies to all listed NGPDU models.

NOTE: Use proper Tag and Lock-Out safety procedures!

If not done previously, shutdown system software and turn off the mains input power to the system.

1.1 UPS Input Power Cable

1) Remove the top and front covers of the PDU.
2) Remove all three jumper wires at terminal strip TS4, terminals 1 to 6, 2 to 7, & 3 to 8. Place the jumpers in the base of the pdu for future use.
3) Route the UPS Input Power Cable (P/N 5169294) into the PDU and up through the indicated cable clamp in the bottom pan. Refer to **Error! Reference source not found.** below.
4) Cut off the RED wire at the cable jacket and set aside the cutoff piece.
5) Connect the BLACK wire to TS4-1
6) Connect the ORANGE wire to TS4-3.
7) Connect the BLUE wire to TS4-4.
8) Connect the GREEN/YELLOW wire to TS4-5.
9) Using the RED wire set aside above, cut to length required and install as a jumper between TS4-2 and TS4-7. Alternatively, you may re-use the original jumper removed in step 2 above.

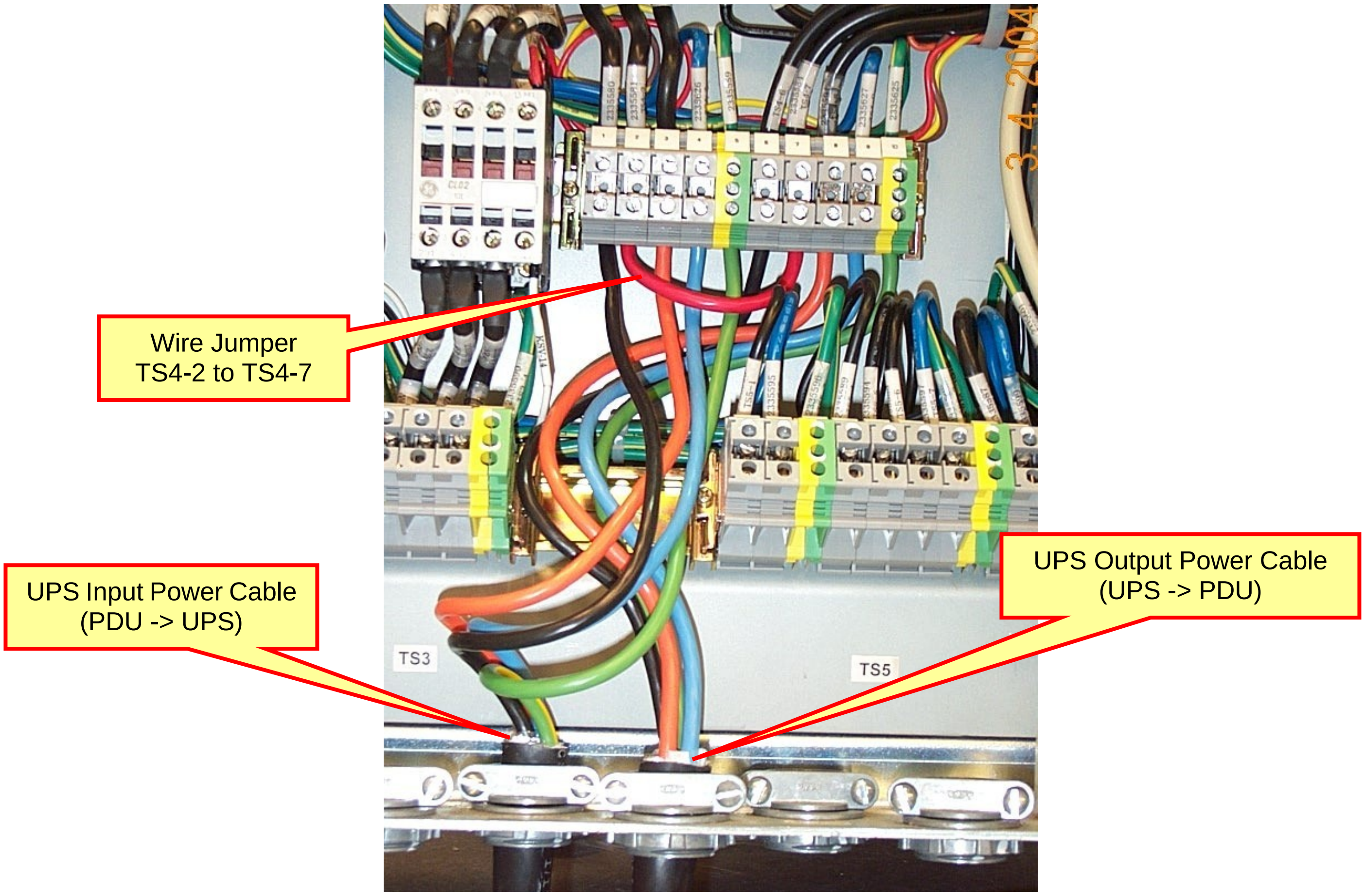


Figure 11 - NGPDU Connections

1.2 UPS Output Power Cable

1) Route the UPS Output Power Cable (P/N 5169294-2) into the PDU and up through the indicated cable clamp in the bottom pan.
2) Cut off the RED wire at the cable jacket and discard.
3) Connect the BLACK wire to TS4-6
4) Connect the ORANGE wire to TS4-8.
5) Connect the BLUE wire to TS4-9.
6) Connect the GREEN/YELLOW wire to TS4-10

Note: Do not re-install the PDU covers at this time. Wail until after the system start-up procedures have been completed.

Section 2 - CT Compact PDU Connections

This UPS kit may be used with older LightSpeed 1.x-4.x CT systems. These systems utilize a CT Compact PDU, model numbers in use are:

- 2133533
- 2269902

The interconnection of the UPS and PDU is similar to the NGPDU interconnect described above. Follow terminal number and detailed interconnection information as shown in the accompanying schematics. Re-label cables as indicated on the schematic appropriate for the PDU model being used.

Chapter 4
A1 Disconnect Panel

Section 1 - A1 Main Disconnect Panel

For safe and orderly system start-up & shutdown this UPS kit must be installed with a properly configured A1 Main Disconnect Panel. Use of one of the following GE panels, or equal, specifically designed for use with this kit, is recommended. Refer to your system Pre-Installation manual for proper selection.

- E4502AB / P5051RJ (90A)
- E4502AC / P5051RK (110A)
- E4502AE (125A)
- E4502AF (150A)

Figure 12 - Typical A1 Panel Front View & Interior View.

DANGER

THE FOLLOWING PROCEDURES ARE ACCOMPLISHED INSIDE THE A1 DISCONNECT PANEL. MAINS VOLTAGE MAY STILL BE PRESENT EVEN WITH THE MANUAL SWITCH PULLED. LOCATE THE FEEDER DISCONNECT TO THE A1 PANEL AND SHUT OFF MAIN POWER THERE. USE PROPER LOCK-OUT / TAG-OUT PROCEDURES. TEST THE A1 DISCONNECT TO VERIFY THAT ALL POWER IS OFF BEFORE PROCEEDING.

1.1 UPS Control Cable Connections to A1 Panel

An electrician will be required to install a 3/4" conduit or raceway between the A1 Disconnect and the floor in the general area of the UPS. Conduit should be terminated in a small wall box. This route will be used for the UPS Control cable connecting the UPS and A1 Main Disconnect Panel.

1) **Verify the feeder to the A1 Panel has been shut off and properly locked out.**

WARNING

REMOVE THE FACTORY INSTALLED JUMPER BETWEEN TERMINALS TB1-7 & TB1-9 IN THE A1 PANEL, IF PRESENT. THIS IS NECESSARY TO PREVENT DAMAGE TO THE UPS EPO CIRCUITRY.

2) Install a ½" cord grip (supplied with the kit) in the wall box or the base of the A1 Panel.
3) Route the UPS Control Cable (P/N 5169224) from the UPS through the conduit or raceway to the A1 Disconnect Panel.
4) Connect the BLK 1 wire to terminal TB1-7 in the A1 Panel.
5) Connect the BLK 2 wire to terminal TB1-8 in the A1 Panel.
6) Connect the BRN wire to terminal TB1-9 in the A1 Panel.
7) Connect the BLU wire to terminal TB1-10 in the A1 Panel.
8) Connect the GREEN/YELLOW (GND) and SHIELD wires to Ground Terminal (GND) in the A1 Panel.

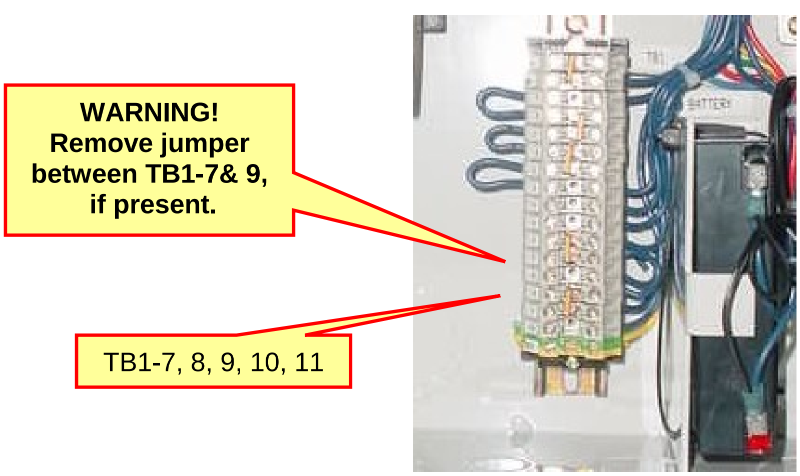


Figure 13 - A1 Panel TB1 Terminals

1.2 A1 Panel Voltage Selection

Verify the A1 Panel Control Transformer is configured for the correct nominal mains voltage. The A1 panel ships from the factory configured for 480V input. Reconnect for 400V if required.

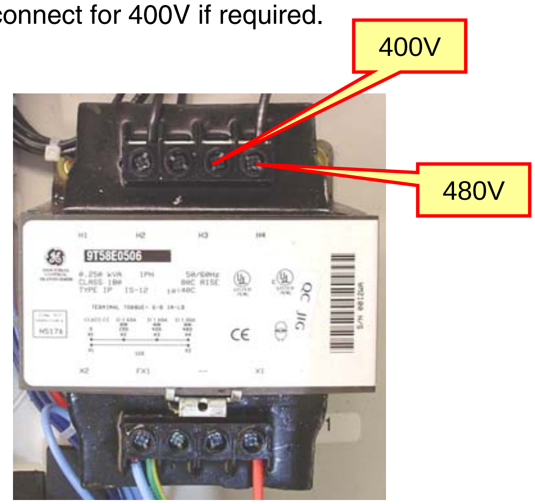


Figure 14 - A1 Panel Control Transformer

1.3 A1 Panel Labeling

A pre-printed adhesive label is provided in the UPS kit titled “SYSTEM OFF PROCEDURE” and “SYSTEM START-UP PROCEDURE”. Attach the label to the front of the A1 Panel as indicated below.

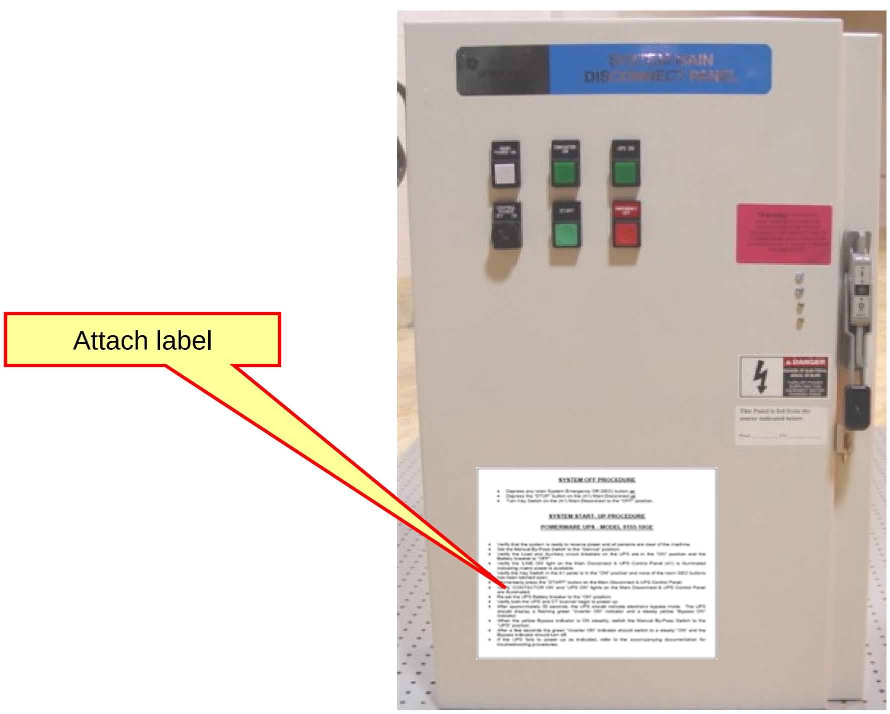


Figure 15 - A1 Panel Labeling

Section 2 – Non-Standard A1 Main Disconnect Panel

Although designed specifically for use with the previously referenced A1 panel catalog items, under certain conditions, this UPS kit may be used with some alternate A1 Panels.

The following configurations are NOT intended for new system installations, but they are listed here to support UPS upgrades / replacements in installed base systems only. Alternate A1 panel configurations include:

- E4502AD (When modified by application of GE Supply kit GESAD2ABKIT1)
- E4502AD or FCA190-A (When coupled to UPS Control Panel 2143207, previously sold as part of UPS kit B7999PS. Note - Field modification of 2143207 required)
- 2385535 (previously sold as part of mobile kit B7999PY)

The interconnection of the UPS and these A1 Panels or UPS Control Panel is not described in detail here. Refer to the detailed interconnection information as shown in the accompanying schematics. Re-label cables as indicated on the schematic appropriate for the configuration being used.

Chapter 5
Start Up / Shutdown Procedures

NOTE:

GEHC installation team and service employees are authorized to perform initial power-up of the Powerware 9155-10GE UPS. It is not necessary to call a Powerware representative for initial start-up. However, if you prefer or have difficulty with the UPS start-up procedure, you may arrange for a Powerware service technician to perform start-up, at no cost. The Powerware contact number in the US is 1-800-843-9433. For international service refer to the contact information provided with the Powerware documentation.

Section 1 - Start-Up: Powerware 9155-10GE

1.1 Automatic Configuration Mode

Automatic configuration is performed during UPS startup. The UPS will automatically sense the input voltage and frequency and match its output to the mains input power.

No additional frequency or voltage configuration set-up is required.

1.2 Auto Re-start

The UPS has been configured at the factory to automatically start-up whenever it senses input power. If the UPS does not automatically start, Powerware service should be contacted to correct the problem and verify proper operation.

1.3 Initial Start-up Sequence

The following procedure should be used to power up the system up for the first time after installing the UPS kit. Verify that all connections are properly made and all safety covers are properly in place. Leave the PDU front cover off. Turn OFF the Console power switch. This keeps the system computers from needlessly booting up until proper UPS system operation is verified.

Note: It is strongly recommended to have the User's Guide for the UPS available.

Check the following before proceeding to power up the system:

- Mains input power was turned OFF during installation. Restore the power at the feeder panel supplying the A1 Disconnect Panel after finishing the A1 connections procedure.

1) Verify that the system is ready to receive power and all persons are clear of the machine.
2) Set the Manual By-Pass Switch to the "Service" position.
3) Verify the Load and Auxiliary circuit breakers on the UPS are in the "ON" position and the Battery breaker is "OFF".
4) Verify the 'POWER ON' light on the Main Disconnect Panel (A1) is illuminated indicating mains power is available.
5) Verify the Key Switch in the A1 panel is in the "ON" position and none of the room SEO buttons has been latched open.
6) Momentarily press the "START" button on the Main Disconnect Panel.

7) Verify ‘CONTACTOR ON’ and “UPS ON” lights on the Main Disconnect Panel are illuminated.
8) Re-set the UPS Battery breaker to the “ON” position.
9) Verify both the UPS and CT scanner begin to power up.
10) After approximately 30 seconds, the UPS should indicate electronic bypass mode. The UPS should display a flashing green “Inverter ON” indicator and a steady yellow “Bypass ON” indicator.
11) When the yellow Bypass indicator is ON steadily, switch the Manual By-Pass Switch to the “UPS” position.
12) After a few seconds the green “Inverter ON” indicator should switch to a steady “ON” and the Bypass indicator should turn off.
13) Turn ON the Console power switch and allow software to boot.

NOTE - The UPS is now providing 120VAC power to the Table, Gantry and Console. In the event of a momentary mains power loss, scanning will be inhibited or aborted. However, the UPS will continue to provide power to the Table, Gantry & Console for at least 10 minutes. The UPS will “beep” occasionally, letting you know that it is on BATTERY BACKUP. If mains power is restored prior to shutting down the CT system, the UPS and system will automatically return to normal operation.

1.4 UPS Alarm Conditions

1.4.1 Input Phase Rotation

If the UPS does not start properly and the display indicates a “**Phase Rotation**” message, there is an input phase error.

Two of the main input phases must be interchanged at the **PDU MAINS INPUT** Terminal Block. Do **NOT** reverse any of the connections at the UPS or PDU TS4 Interface.

1.4.2 Other Alarms

Refer to the 9155-10GE User’s Guide for a detailed listing of operational and diagnostic information for the UPS.

Section 2 - Shut-Down Procedure

2.1 Normal System Shutdown

To shut down the entire system, do the following:

1) Perform a software shutdown.
2) Turn off the Console power switch.
3) Press any room System Emergency Off (SEO) button or
4) Press the STOP button on the A1 Main Disconnect Panel or
5) Set the manual switch of the A1 Disconnect Panel in the OFF position.

To re-start the system, follow the Start-Up instructions in the previous section.

2.2 UPS Shutdown Only – Manual Bypass

To shut down the UPS only while leaving the system operational, do the following:

1) Turn the Manual Bypass switch to the BYPASS position.
2) The UPS should automatically shut down and trip its battery breaker.
3) Verify the UPS battery circuit breaker CB1 is tripped to the “OFF” position.

Comment:

Although the UPS stops supplying power and is disconnected internally from its batteries, system power still passes through the UPS cables while in bypass mode. Use caution when servicing.

2.3 Final Steps

1) Re-install all safety shields and covers on the UPS and PDU.
2) Fill out and mail the Powerware Warranty Registration card or register the UPS on-line at www.warranty.powerware.com.

Appendix A
Applicable Documents

The following documents provide reference information for the UPS Interconnect Kit.

Section 1
GEHC Reference Documents

- 5169128SCH	UPS Interconnect Schematic
- 5169350PSP	2Ph, 10kVa UPS Purchase Specification
- 2326492SCH	NGPDU Schematic
- 2391993PSP	Purchase Specification for A1 Main Disconnect & UPS Control Panel
- 5169123ADW	Assembly Specification for the UPS Hardware Kit
- 46-015900	Global Documentation Standard
- 46-131001	Applied Practices (Assemblies)
- 46-170700	Applied Practices (Portions)

Section 2
Vendor Documents

- POWERWARE 9155-10GE User's Guide

Section 3
Industry Standards

- NFPA-70 (2005) National Electrical Code (USA)

Appendix B
Field Replaceable Unit (FRU) Listing

The following listing identifies the Field Replaceable Units (FRUs) associated with this assembly. Vendor part numbers are included as examples only. For the complete list of approved vendors for each item, see the associated Manufacturer's Equivalent on GEHC ePDM system.

GEMS PART #	NAME / DESCRIPTION	VENDOR	VENDOR P / N
5169350	PW9155-10GE UPS	POWERWARE	G410110000
5169123	Hardware Kit	GE Supply	5169123

Figure 16 - Field Replaceable Units

GE Healthcare
GE HEALTHCARE-AMERICAS: FAX 262.312.7434
3000 N. GRANDVIEW BLVD., WAUKESHA, WI 53188 U.S.A.